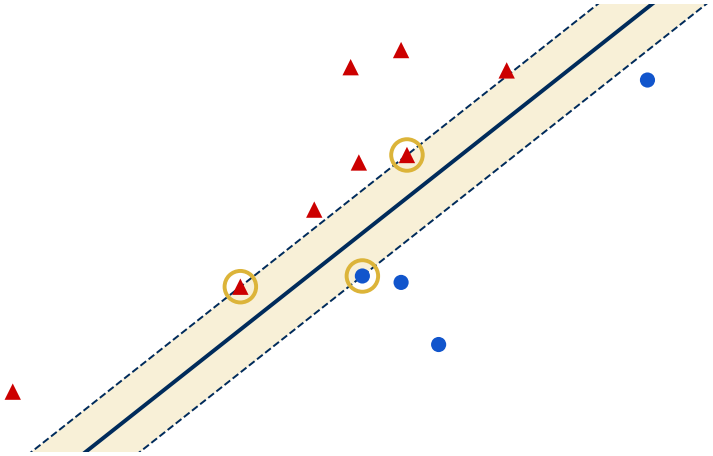
UNIT 02 · CHAPTER 9

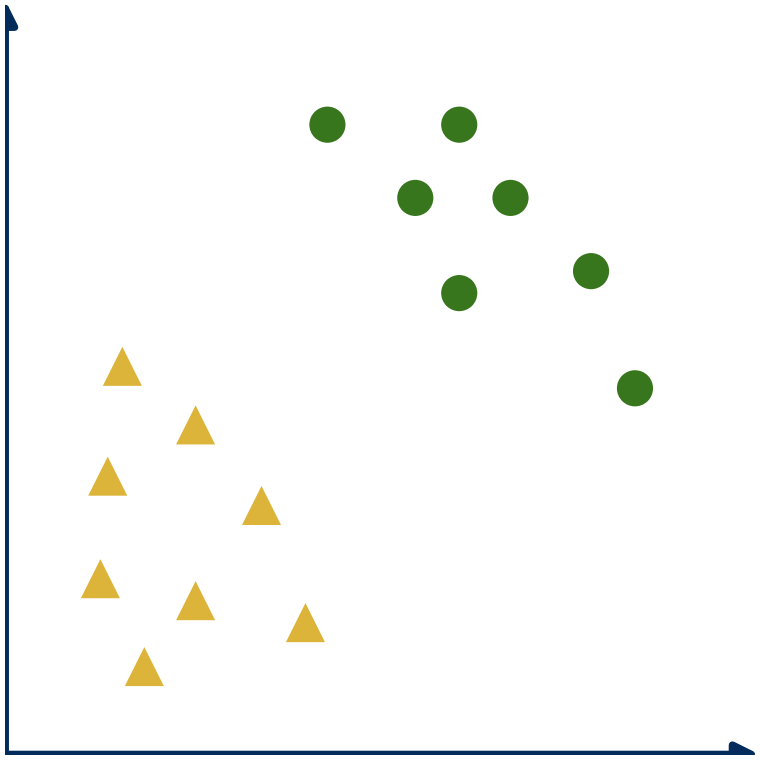
# Support Vector Machines



CSE 30124 · Introduction to Artificial Intelligence
William Theisen · University of Notre Dame

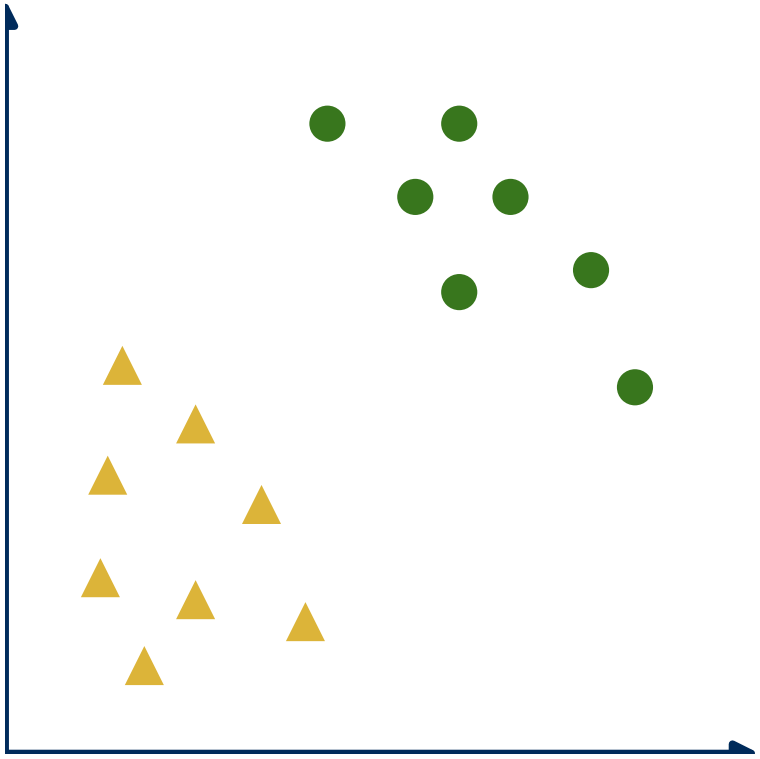
# Linear Classifiers (exercise)

Given the dataset below, **where would a Perceptron draw a separating line?**

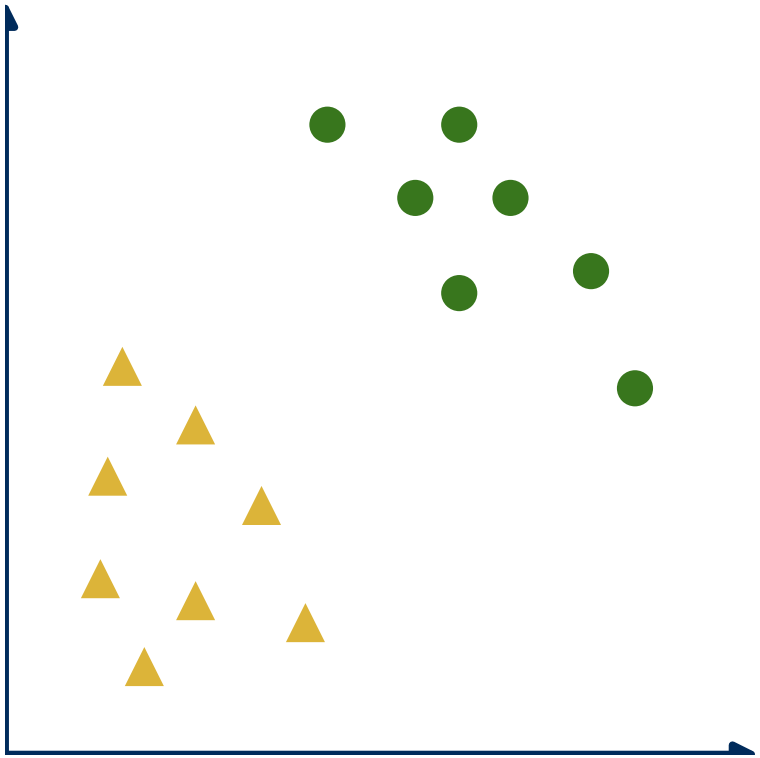


# Linear Classifiers

Given the dataset below, **where would a Logistic Regression draw a separating line?**

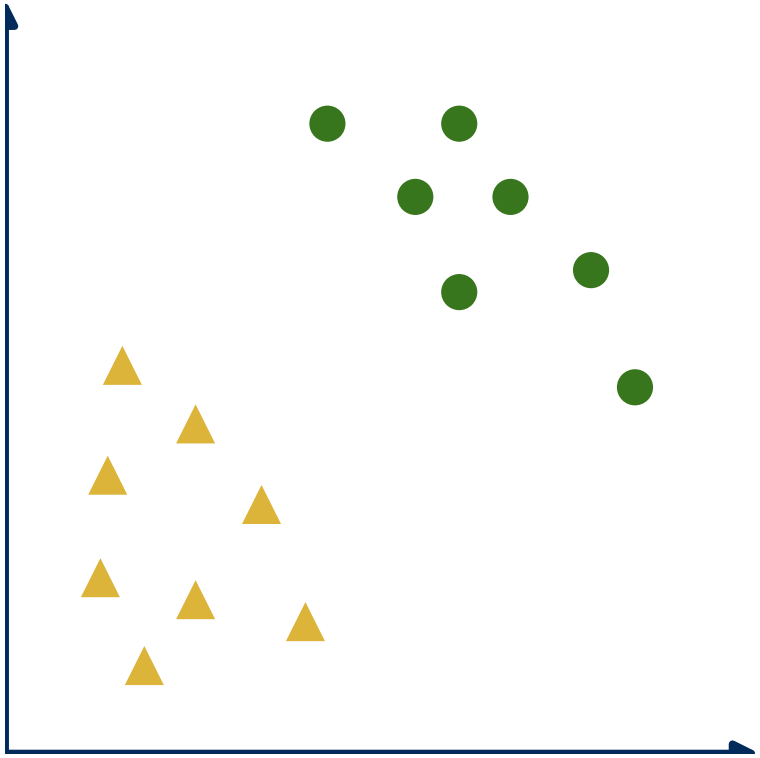


# Margin Classifier



# Support Vectors

**Support Vectors** are the points ___ or _____ the **margin** that define the **decision boundary**



# Support Vector Machines (SVMs)

**Support Vector Machines** are a kind of **Binary Classifier** that focus on the concept of a _____ to make more robust and efficient classifiers

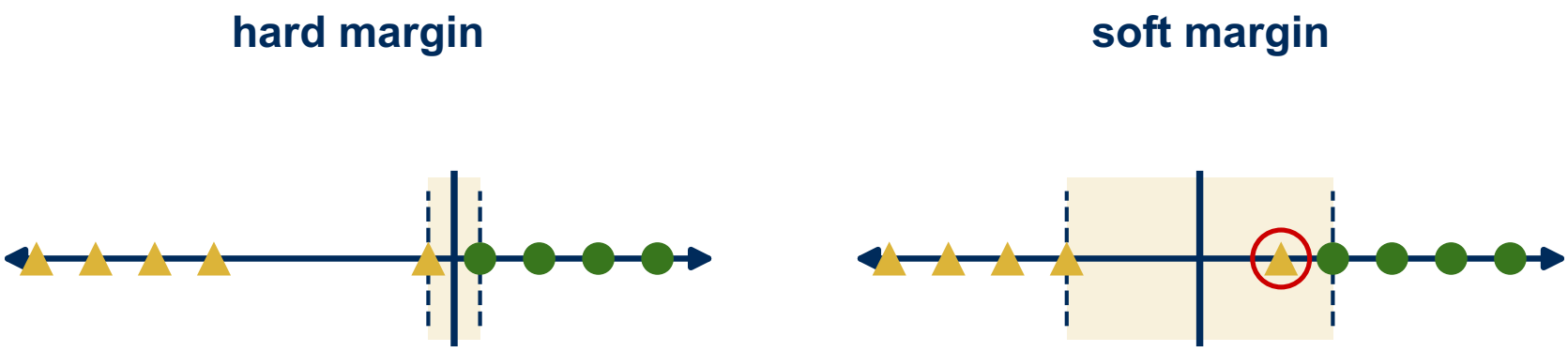
# SVMs in Biology

Golub et al. (1999) measured the activity of about 7,000 genes in samples from 72 leukemia patients and asked: can we tell the two main types (AML and ALL) apart from the genes alone?

That's **thousands of features and only dozens of samples**, so infinitely many lines separate the training data perfectly and most of them are bad.

Picking the line with the **largest** _____ is a sensible way to choose, and SVMs (Furey et al., 2000) became one of the standard tools for **classifying tumors from gene activity**.

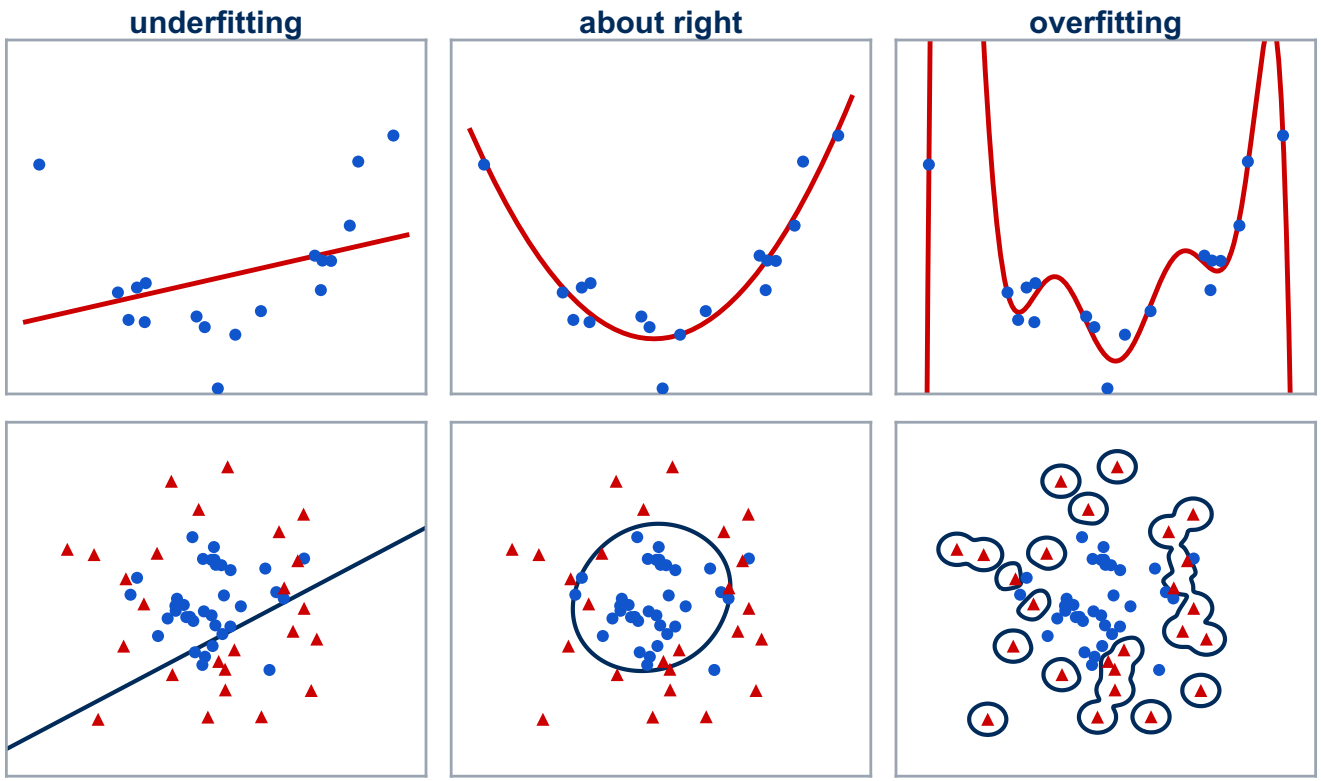
# Hard-Margin vs. Soft-Margin SVMs




**Hard-Margin SVMs allow no misclassifications**, much like a Perceptron

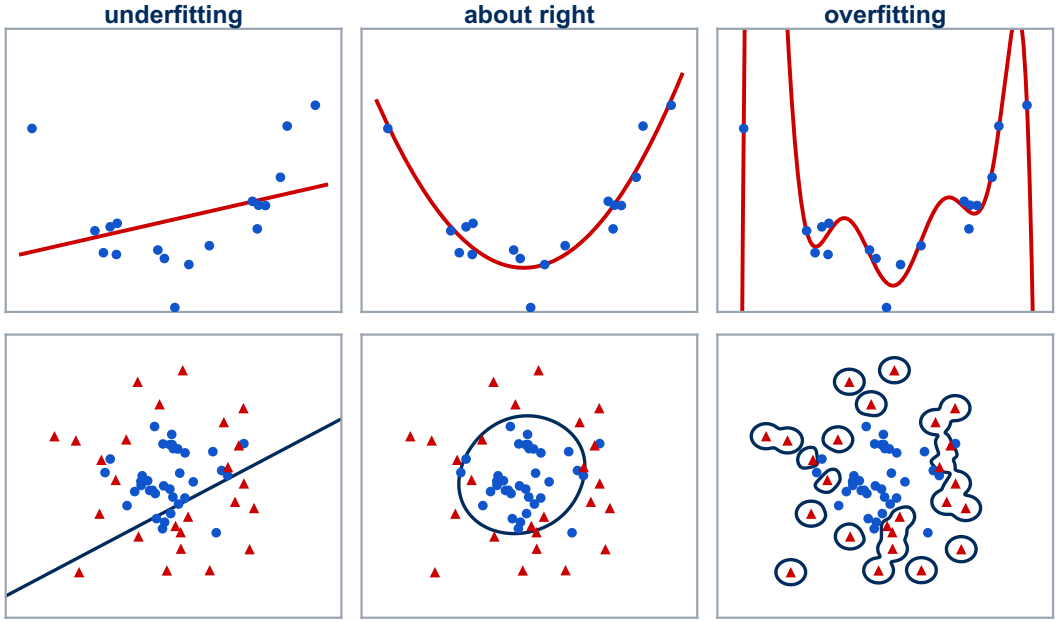
**Soft-Margin SVMs allow for some misclassifications** to allow for a more robust decision boundary

# Underfitting and Overfitting



# The bias-variance tradeoff (reading)

**Underfitting** and **overfitting** are _____ of having a **high bias** or a **high variance**



# The bias-variance tradeoff (reading)

"the **bias–variance** tradeoff describes **the relationship between a model's complexity, the accuracy of its predictions during training, and the accuracy of its predictions during testing.**" (Modified from Wikipedia)

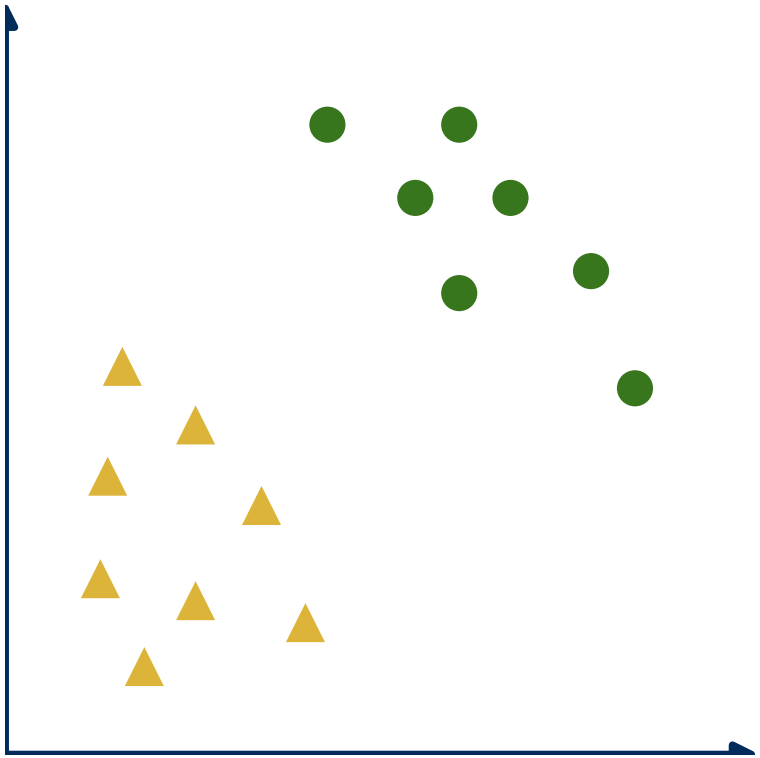
# The bias-variance tradeoff (reading)



# The bias-variance tradeoff (reading)

| High Bias (Low Variance) | High Variance (Low Bias) |
| --- | --- |
| _____ | _____ |
| _____ | _____ |
| _____ | _____ |
| _____ | _____ |
| _____ | _____ |

# Hard-Margin Classifier Definition



# Hard-Margin Classifier Definition

We want to **minimize ||ω||** such that

**all points are correctly classified** and

**all points are on or outside of the margin**

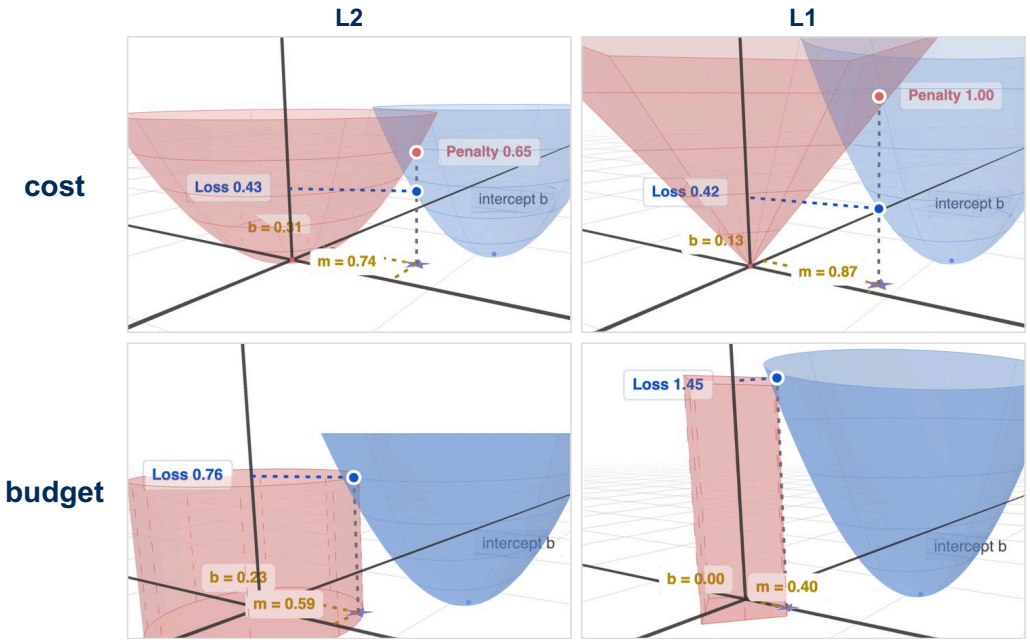
# Lagrangian Multipliers

A strategy for finding the **local** maxima and **minima** of a **function subject to constraints**

Essentially, allows us to wrap the "s.t." into a **single cost/error/loss function**

# Regularization

**Lagrangian Multipliers** are also how regularization works

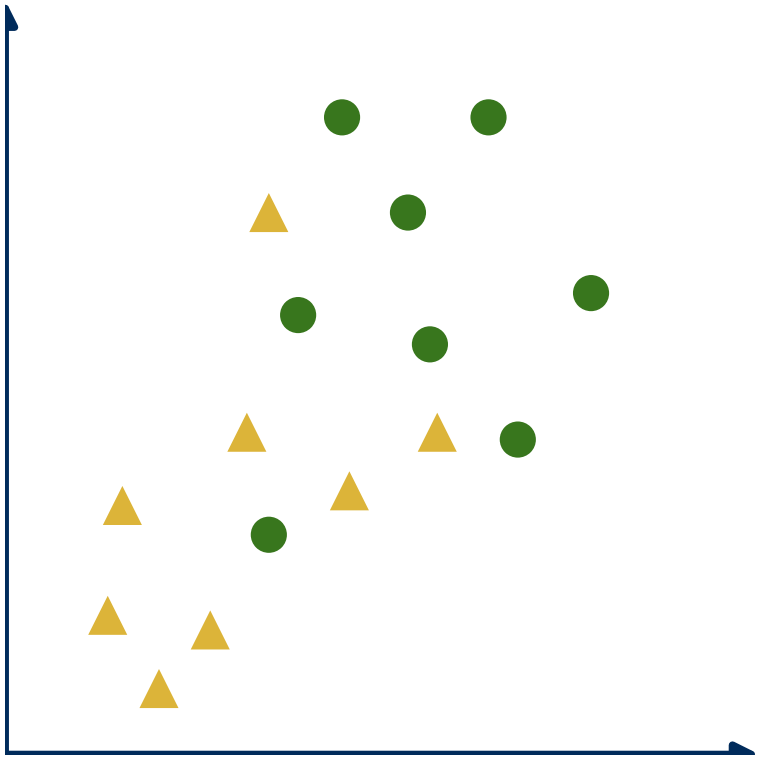



They allow us to transform the "s.t." into a **single cost/error/loss function**

# Hard-Margin SVM Optimization (math)

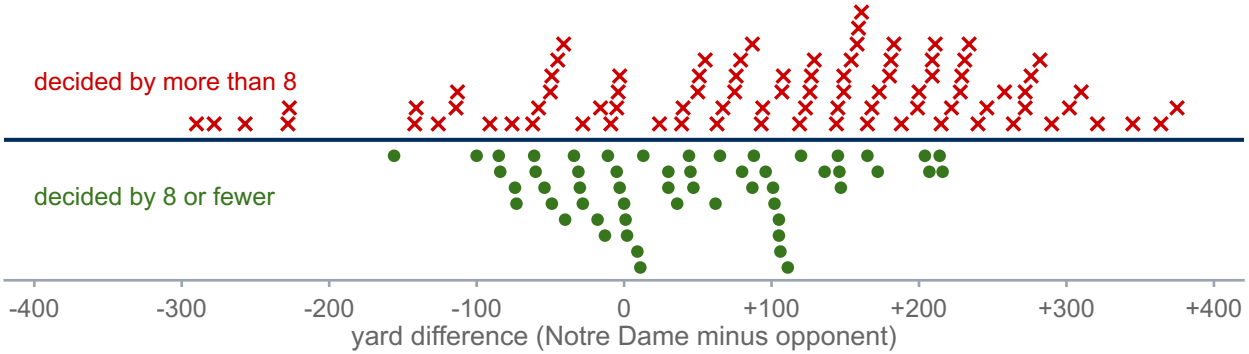
# Linear Classifiers (exercise)

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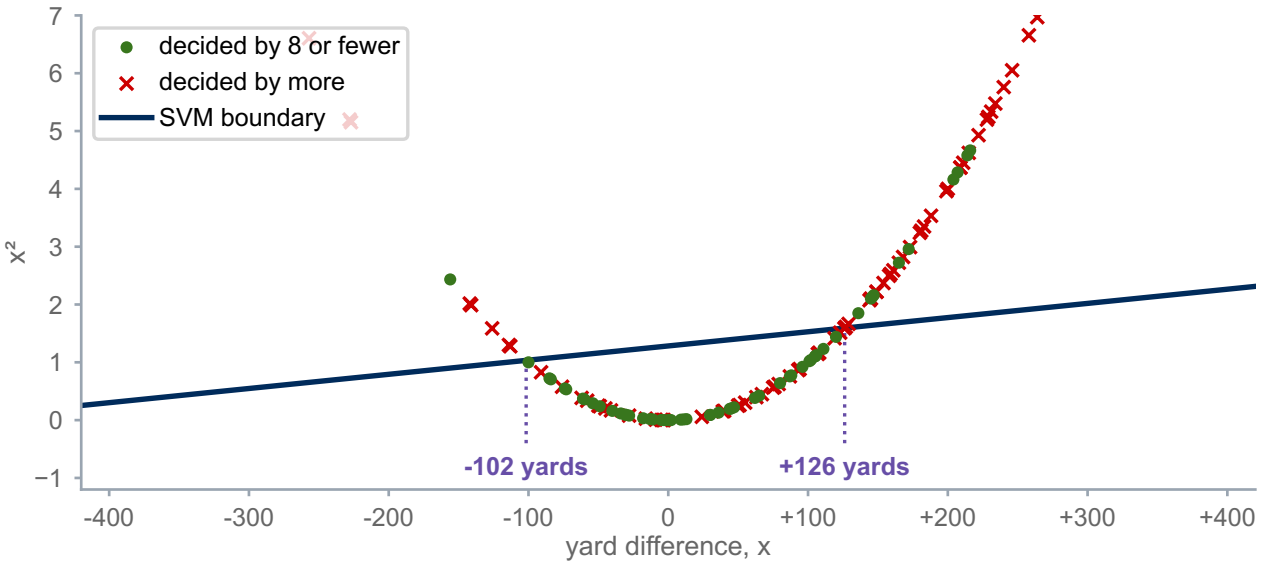


# Soft-Margin SVMs (math)

# Non-Linearly Separable Data (reading)



# Non-Linearly Separable Data (reading)

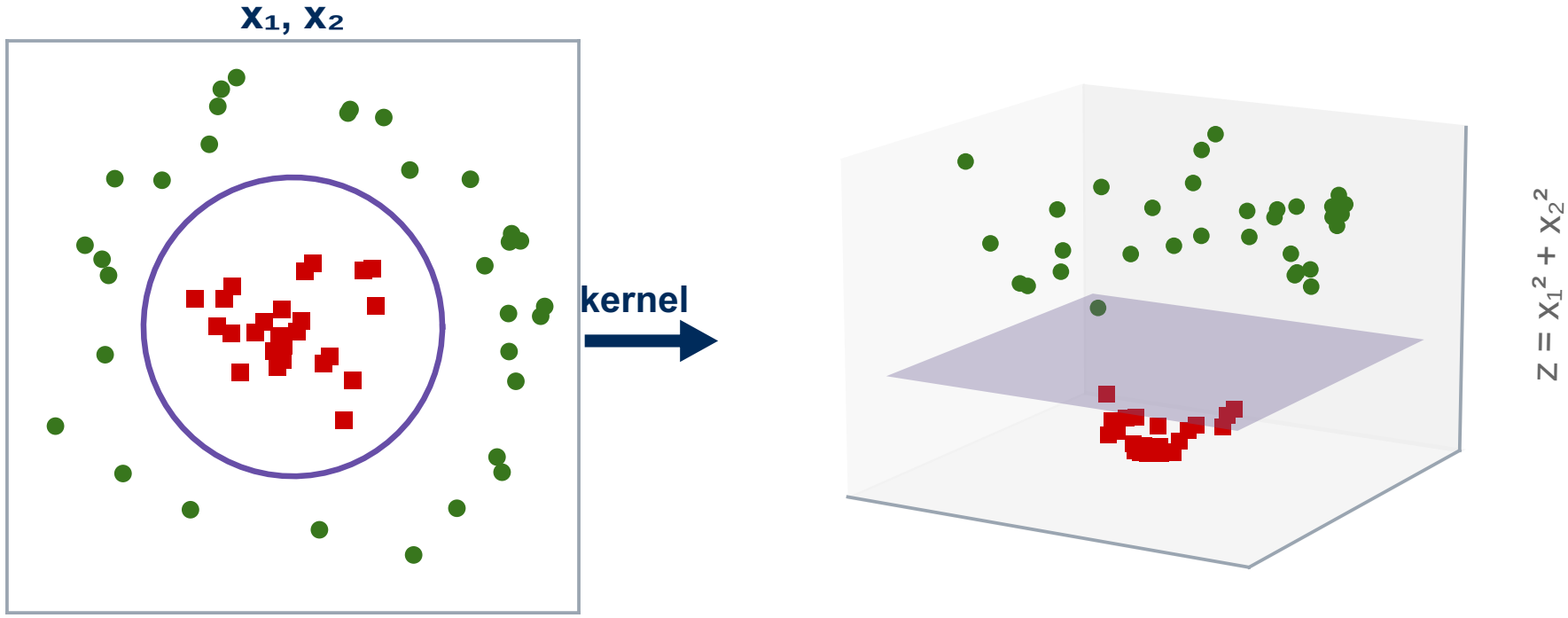



Changing your space is a very powerful idea!

# The Dual-Form Equation

# The Kernel Trick

What makes **SVMs** so powerful is their ability to implicitly _____ _____ via the **"Kernel Trick"**

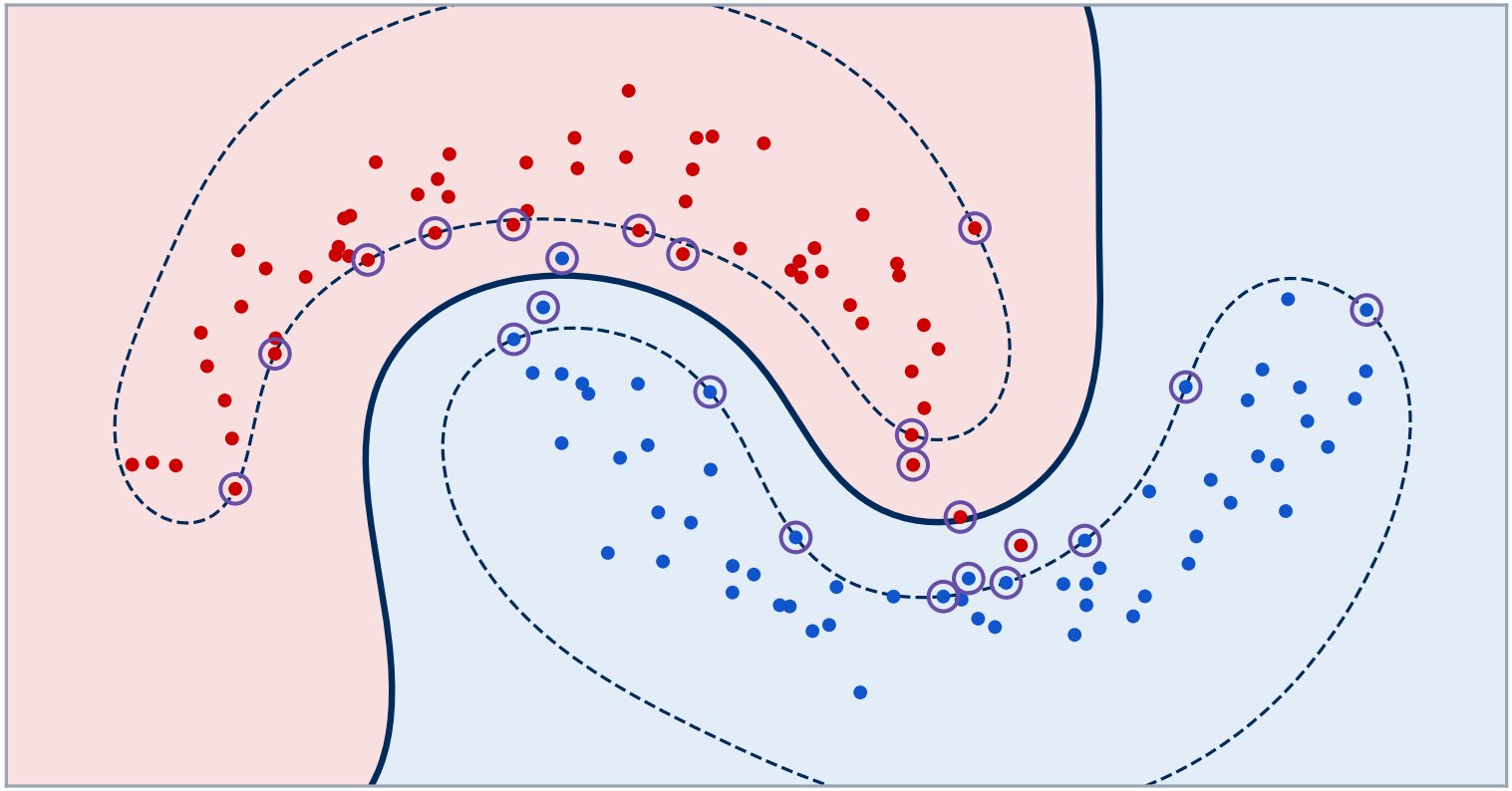


# Common Kernels

There are **three common kernels**:

- _____
- _____
- _____

# Visualization



# Matching: Support Vector Machines

| | |
|---|---|
| _____ **Margin** | **A.** Allows some misclassifications for a more robust decision boundary |
| _____ **Support vectors** | **B.** The gap between the decision boundary and the closest points of each class |
| _____ **Hard-margin SVM** | **C.** Allows no misclassifications, much like a Perceptron |
| _____ **Soft-margin SVM** | **D.** Fast and simple, but only finds a straight-line (hyperplane) boundary |
| _____ **Lagrangian multipliers** | **E.** Builds very complex boundaries from distances; needs tuning of γ and C |
| _____ **Kernel trick** | **F.** Computes dot products in a higher-dimensional space without transforming the data |
| _____ **Linear kernel** | **G.** Finds curved boundaries of a chosen degree; can overfit at high degrees |
| _____ **Polynomial kernel** | **H.** The points on or inside the margin that define the decision boundary |
| _____ **RBF kernel** | **I.** Wrap the "s.t." constraints into a single cost/error/loss function |

# Practice problems

**1.** What makes an SVM different from a Perceptron with respect to the decision boundary?

**2.** What are support vectors, and why are they important in SVM?

**3.** When may we prefer a soft-margin classifier over a hard-margin classifier?

# Practice problems

**4.** How does SVM handle non-linearly separable data? What role does the kernel function play in this?

**5.** What are the three most common kernels used in SVMs?

**6.** What is the kernel trick? (1 pt.)

# Practice problems

**7.** What happens if you were to change the location of a support vector? Is it different depending on if we're using a hard-margin or soft-margin classifier? (2 pts.)